

Assignment

Part 1 — Project Update (1–2 pages)

Briefly revisit your original proposal.

Answer:

1. **Current Project Focus**

- In one clear paragraph: What exactly are you trying to investigate?
- Has your focus changed since Sprint 1?

1. **What You Now Understand Better**

- List 3–5 important ideas, terms, or concepts that are central to your project.
- Briefly explain each in your own words.

1. **What Is Still Unclear**

- What confuses you?
- What technical or conceptual risks do you see?

Keep this section concise and concrete.

Part 2 — Literature and Resource Review (1–3 pages)

You will review and synthesize material related to your project.

You may use:

- Documentation (R manuals, package documentation)
- Books (e.g., *Advanced R*, *The Book of R*, etc.)
- Academic papers
- Technical articles
- Blogs
- Videos
- Tutorials
- Benchmark case studies
- GitHub repositories

There is no required number of each type of source.

However:

- Your review must show that you looked at *multiple perspectives*.
- All sources must be properly cited.

- Your reference list must include everything you used (including videos and blogs).
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How to Write This Section

Do **not** summarize sources one by one.

Instead:

1. Identify 3–5 important themes that appear across what you read.
2. Organize your review around those themes.
3. For each theme:
 - Explain what the general idea is.
 - Explain what different sources say.
 - Identify any disagreements or tradeoffs.
 - State what seems most convincing to you.

Examples of possible themes (do not just copy these — think yourself):

- Performance vs readability
- Abstraction vs control
- Vectorization vs iteration
- Database-side computation vs in-memory computation
- Reliability of benchmarks
- Code maintainability

Your job is to decide what matters for your project.

Part 3 — What You Plan to Test (1–2 pages)

Based on your reading, clearly state:

1. One or Two Questions You Want to Answer

Examples:

- Is `apply()` faster than a `for` loop?
- Does running aggregation in SQL reduce memory usage in R?
- Are very small benchmarks reliable?

Make the question precise.

2. What Would Count as Evidence?

Explain:

- What you would measure
- What comparison you would run
- What result would support your claim
- What result would contradict it

You do not need to run experiments yet.

But you must think clearly about what “proof” would look like.

Part 4 — **AI Use Disclosure (Required)**

You are allowed to use AI-assisted tools (e.g., ChatGPT, Copilot, Gemini, Claude, etc.) for idea generation, clarification, and drafting support. However:

- You remain responsible for correctness.
 - You must understand and be able to explain everything you submit.
 - You must disclose how you used AI.
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You Must Include the Following

1. Tool Identification

List:

- Name of tool(s) used
- Date(s) used
- Version (if known)

Example format:

Tool: ChatGPT (GPT-4)

Dates used: March 1–3

Purpose: Clarifying benchmarking pitfalls

2. Exact Use Description

Clearly describe:

- What you asked the AI to do
- What prompts you gave (summarize or quote key prompts)
- What parts of your submission were influenced by AI

Be specific. For example:

- “I asked AI to explain why microbenchmarks below 1 ms may be unreliable.”
- “I asked AI to suggest possible performance metrics.”
- “AI helped restructure my outline.”

Do not write: “I used AI for help.”

3. Verification and Modification

Explain:

- What you changed after receiving AI output
- What you rejected
- How you verified correctness

Examples:

- “I tested the example code before including it.”
- “I cross-checked performance claims against R documentation.”
- “I removed a suggestion because it contradicted package documentation.”

4. Reflection (Short Paragraph)

Answer:

- Did AI improve your understanding?
- Did it confuse you anywhere?
- Where did you rely on your own reasoning instead?

Keep this honest and specific.

Citation Requirement

If AI provided:

- Code
- Explanations

- Definitions
- Structural ideas

It must be cited in your reference list.

Example citation format (choose consistent style):

OpenAI. ChatGPT response to author query on benchmarking R code. March 2, 2026.

If you used multiple sessions, you may group them but must describe them clearly.

Important

Failure to disclose AI use is treated as misrepresentation under the course AI policy.

Using AI transparently and thoughtfully is acceptable.

Using it without understanding or disclosure is not.

Why This Section Exists

This course expects you to:

- Use tools critically
- Retain responsibility for correctness
- Develop independent reasoning

This section ensures intellectual ownership.

Formatting and Submission

Create a Google Docs file.

Include:

- Title
- Your name
- Date
- Clear section headings
- A properly formatted reference list (any consistent style)

Upload to:

2nd Sprint Review

Share link in Canvas (Shared with Lyon College).

Evaluation Criteria (Project Component)

Based on the rubric in the syllabus:

- Clarity of thinking
- Depth of understanding
- Ability to explain technical ideas
- Quality of synthesis (not summary)
- Clear plan for next sprint

This is not a research paper.

It is evidence that you understand your project well enough to implement it thoughtfully.

Expected Time

Approximately 8–12 hours total:

- 4–6 hours reading/watching
- 4–6 hours writing and organizing